

Objectives

- Explain the need and benefits of React Router
- Identify the Components in React Router
- List the types of Router Components
- Parameter passing via url

In this hands-on lab, you will learn how to:

- Implement a Simple Navigation Menu
- Add Basic Routes (install, configure)
- Use Routes in React Applications

Prerequisites

The following is required to complete this hands-on lab:

- Node.js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **60 minutes.**

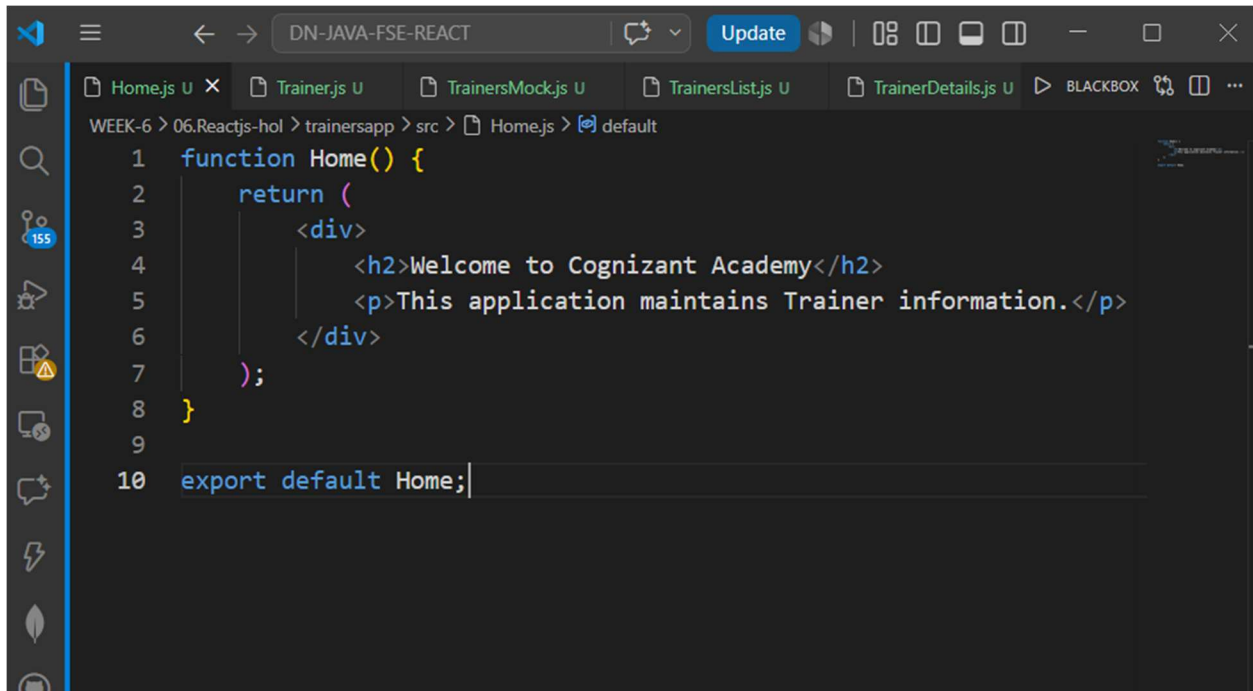
Cognizant Academy teams want to maintain a list of trainers along with their expertise in a SPA using React as the technology. You are assigned the task of creating this React app.

The following trainers' data application will deal.

1. T-ID
 2. Name
 3. Phone
 4. Email
 5. Stream
 6. Skills
-
1. Create a new React app using *create-react-app* tool with the as "TrainersApp"
 2. Open the application using the VS Code
 3. Add a new file called *trainer.js* inside the **src folder** and define a class named as "Trainer" with the following properties
 - a. TrainerId
 - b. Name
 - c. Email

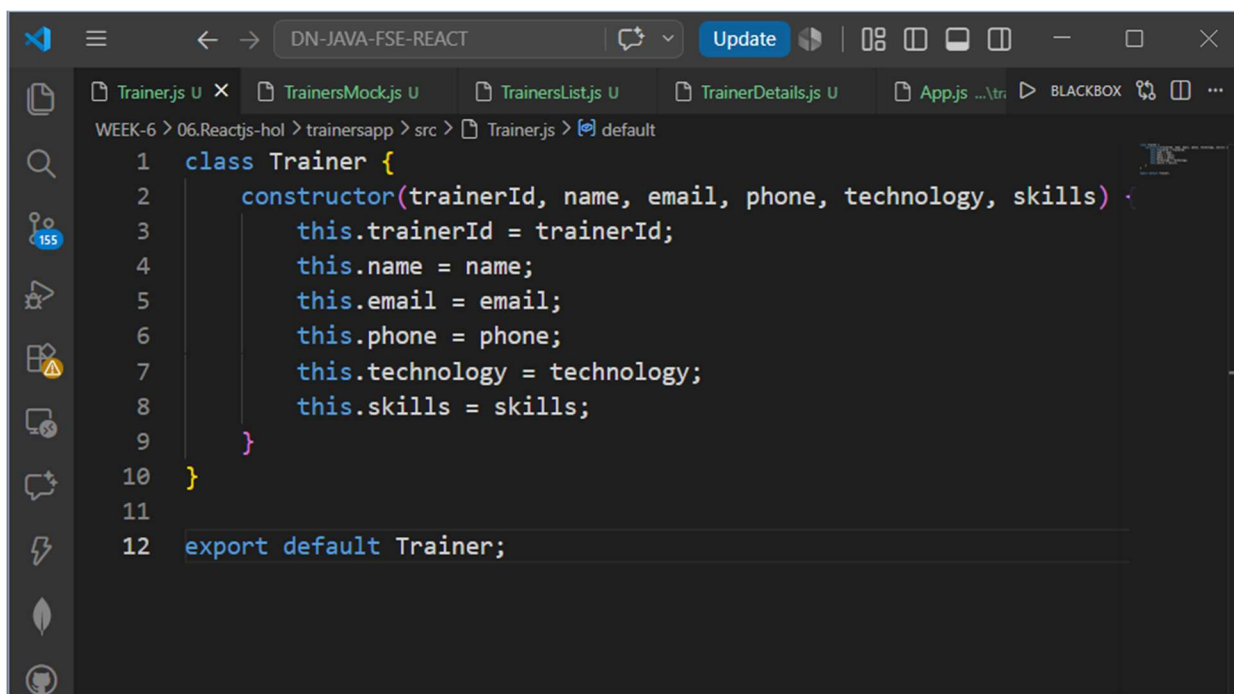
- d. Phone
 - e. Technology
 - f. Skills
4. Create a new `TrainersMock.js` file which will contain the mock trainer data. Refer the following screenshot for mock data
 5. Install the support for React router for the dom. Execute the following command.
 6. Create new component named as **TrainersList** inside *Trainerlist.js* file. The component should accept the trainer's data as parameter and render it as a list. The list should display names of each trainers which must be clickable like a hyper link. Refer the following screenshot for the component layout.
 7. Create a new component named as Home inside `Home.js` which will be responsible for displaying the following
 8. Modify the App component to add support for routing and defining the navigation links to Home component and TrainersList component. Use `BrowserRouter`, `Routes`, `Route` and `Link` components from the `react-router-dom` library.
Define the following URL
 1. `/` - must redirect to home component
 2. `/trainers` – must redirect to trainers list component.The layout of the page must be similar to the following
 9. Create a new component named **TrainerDetail** in *TrainerDetails.js* file.
The component should retrieve a parameter named `id` from the URL with the help of “`useParams`” hook from the React router DOM library.
It should query the mock trainer data using the `id` and display the trainer details as show in screenshot.
Modify the TrainersList component to add Links to TrainerDetail component while passing the ID. Define a route in App component for the same.
 10. Build and run the application. The complete layout of the application will look as follows.

Home.js



```
WEEK-6 > 06.Reactjs-hol > trainersapp > src > Home.js > default
1  function Home() {
2      return (
3          <div>
4              <h2>Welcome to Cognizant Academy</h2>
5              <p>This application maintains Trainer information.</p>
6          </div>
7      );
8  }
9
10 export default Home;
```

Trainers.js

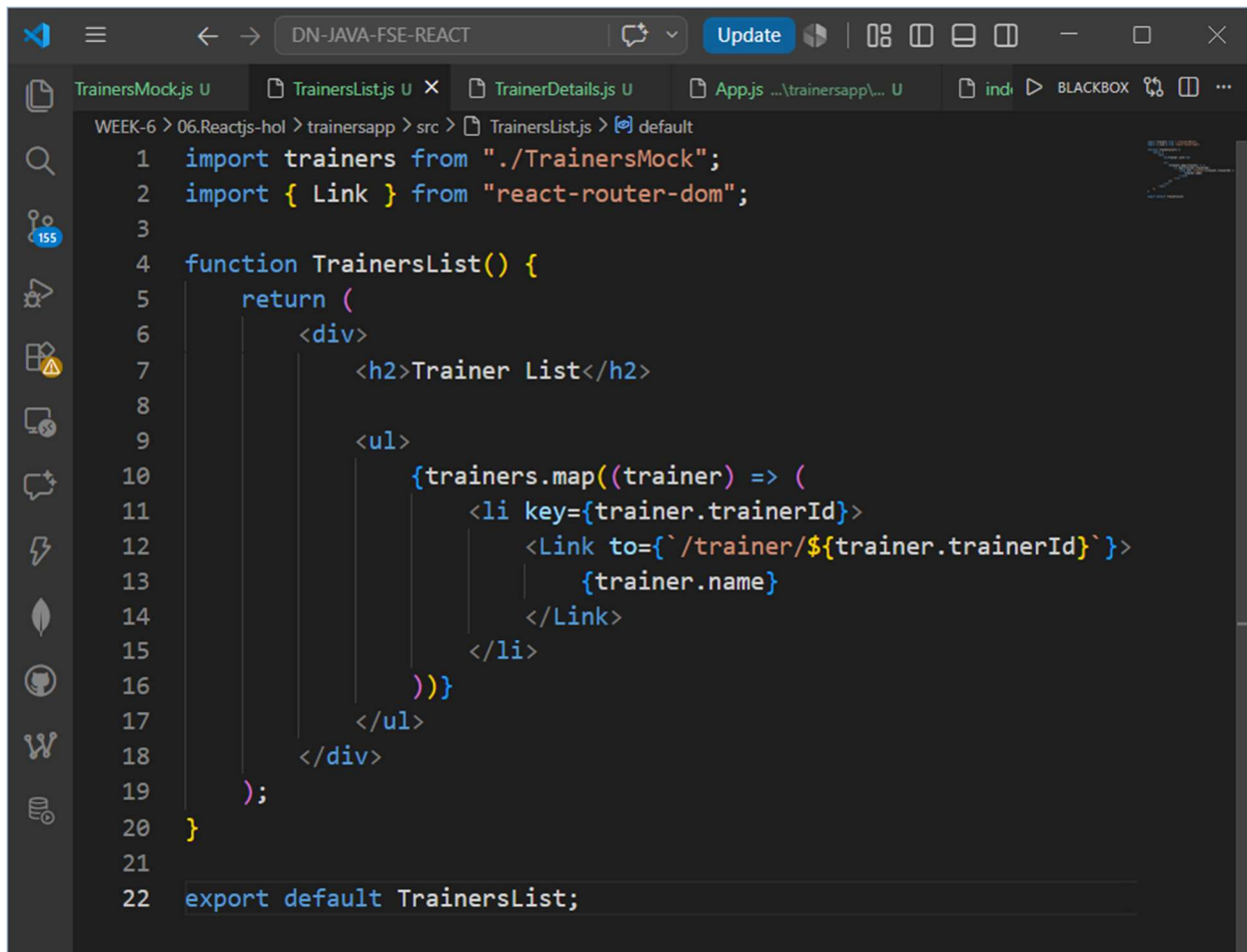


```
WEEK-6 > 06.Reactjs-hol > trainersapp > src > Trainer.js > default
1  class Trainer {
2      constructor(trainerId, name, email, phone, technology, skills) {
3          this.trainerId = trainerId;
4          this.name = name;
5          this.email = email;
6          this.phone = phone;
7          this.technology = technology;
8          this.skills = skills;
9      }
10 }
11
12 export default Trainer;
```

TrainersDetails.js

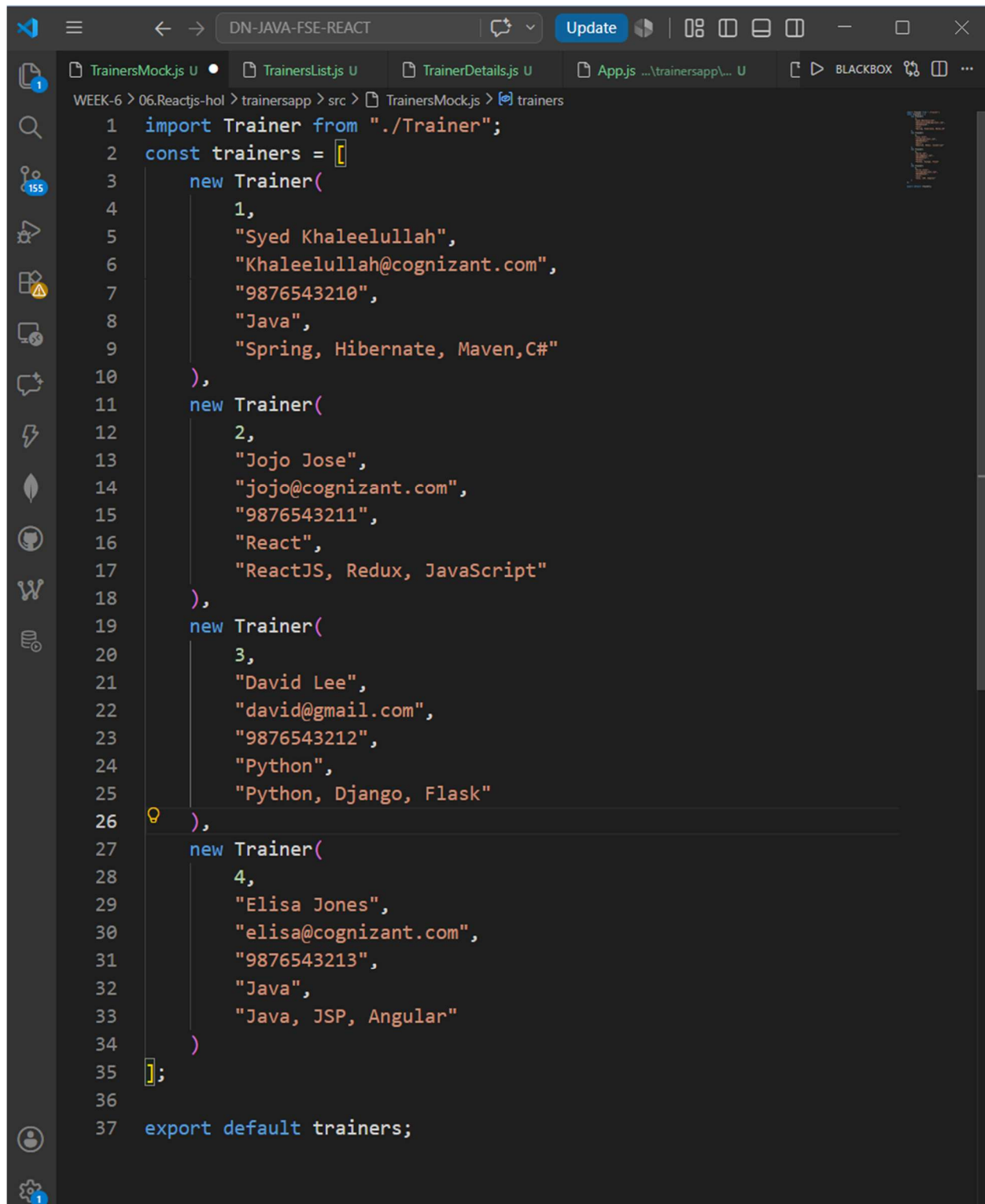
```
WEEK-6 > 06.Reactjs-hol > trainersapp > src > TrainerDetails.js > default
1 import { useParams } from "react-router-dom";
2 import trainers from "../TrainersMock";
3
4 function TrainerDetails() {
5
6     const { id } = useParams();
7
8     const trainer = trainers.find(
9         (t) => t.trainerId === Number(id)
10    );
11
12    if (!trainer) {
13        return <h3>Trainer Not Found</h3>;
14    }
15
16    return (
17        <div>
18            <h2>Trainer Details</h2>
19
20            <p><b>ID:</b> {trainer.trainerId}</p>
21
22            <p><b>Name:</b> {trainer.name}</p>
23
24            <p><b>Email:</b> {trainer.email}</p>
25
26            <p><b>Phone:</b> {trainer.phone}</p>
27
28            <p><b>Technology:</b> {trainer.technology}</p>
29
30            <p><b>Skills:</b> {trainer.skills}</p>
31
32        </div>
33    );
34 }
35
36
37 export default TrainerDetails;
```

TrainersList.js



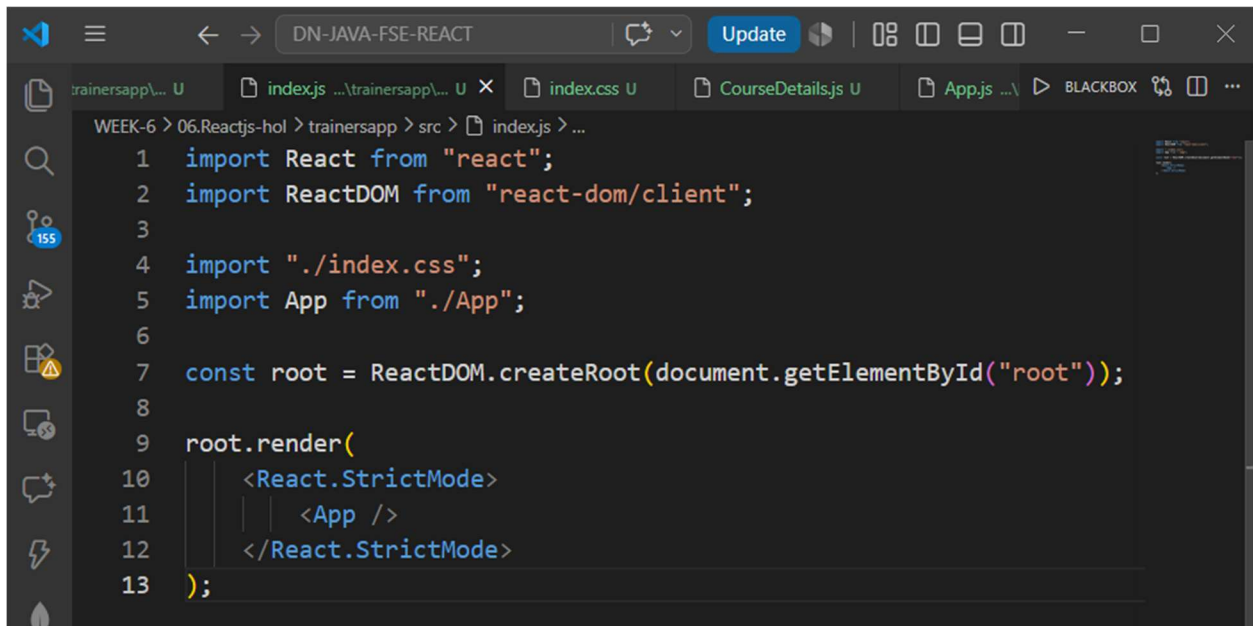
```
WEEK-6 > 06.Reactjs-hol > trainersapp > src > TrainersList.js > default
1 import trainers from './TrainersMock';
2 import { Link } from 'react-router-dom';
3
4 function TrainersList() {
5   return (
6     <div>
7       <h2>Trainer List</h2>
8
9       <ul>
10        {trainers.map((trainer) => (
11          <li key={trainer.trainerId}>
12            <Link to={`/${trainer}/${trainer.trainerId}`}>
13              {trainer.name}
14            </Link>
15          </li>
16        )}}
17      </ul>
18    </div>
19  );
20 }
21
22 export default TrainersList;
```

TrainersMock.js



```
1 import Trainer from "./Trainer";
2 const trainers = [
3   new Trainer(
4     1,
5     "Syed Khaleelullah",
6     "Khaleelullah@cognizant.com",
7     "9876543210",
8     "Java",
9     "Spring, Hibernate, Maven, C#"
10  ),
11  new Trainer(
12    2,
13    "Jojo Jose",
14    "jojo@cognizant.com",
15    "9876543211",
16    "React",
17    "ReactJS, Redux, JavaScript"
18  ),
19  new Trainer(
20    3,
21    "David Lee",
22    "david@gmail.com",
23    "9876543212",
24    "Python",
25    "Python, Django, Flask"
26  ),
27  new Trainer(
28    4,
29    "Elisa Jones",
30    "elisa@cognizant.com",
31    "9876543213",
32    "Java",
33    "Java, JSP, Angular"
34  )
35 ];
36
37 export default trainers;
```

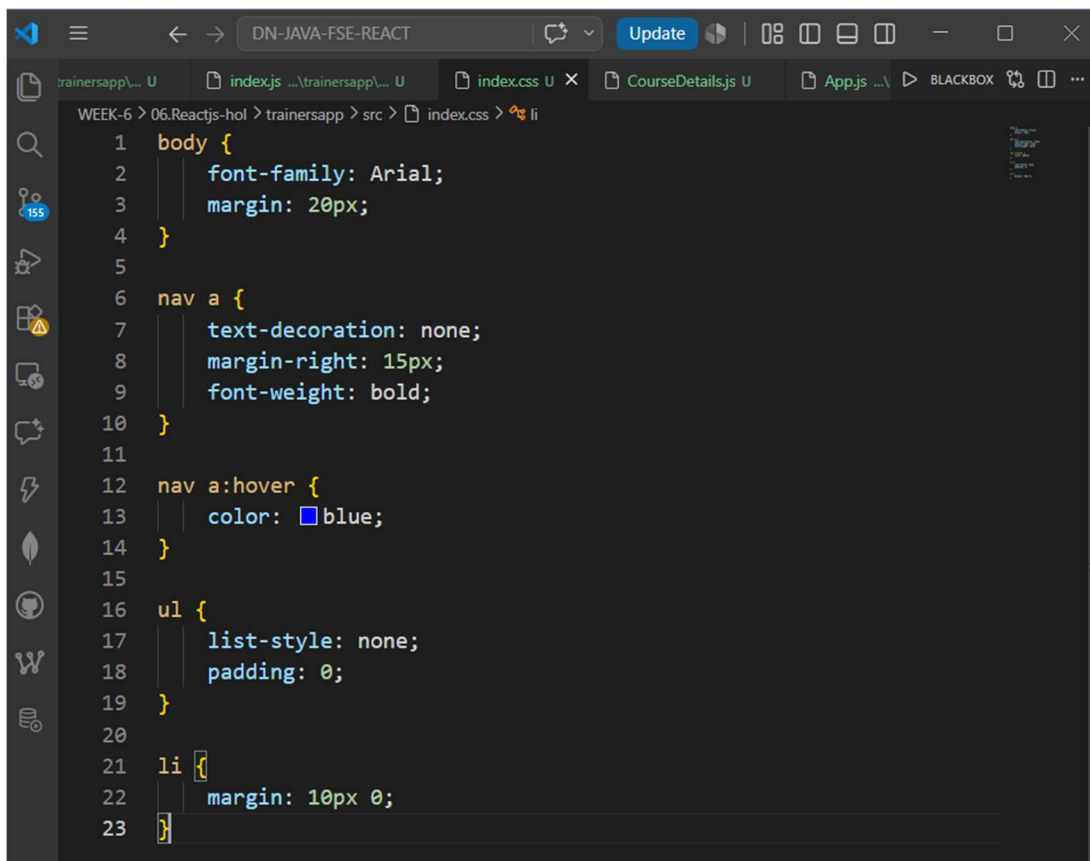
Index.js



The screenshot shows the Visual Studio Code editor with the file `index.js` open. The editor has a dark theme and a sidebar on the left with icons for Explorer, Search, Run and Debug, Extensions, and Testing. The top bar shows the file path `WEEK-6 > 06.Reactjs-hol > trainersapp > src > index.js > ...`. The code in the editor is as follows:

```
1 import React from "react";
2 import ReactDOM from "react-dom/client";
3
4 import "./index.css";
5 import App from "./App";
6
7 const root = ReactDOM.createRoot(document.getElementById("root"));
8
9 root.render(
10   <React.StrictMode>
11     <App />
12   </React.StrictMode>
13 );
```

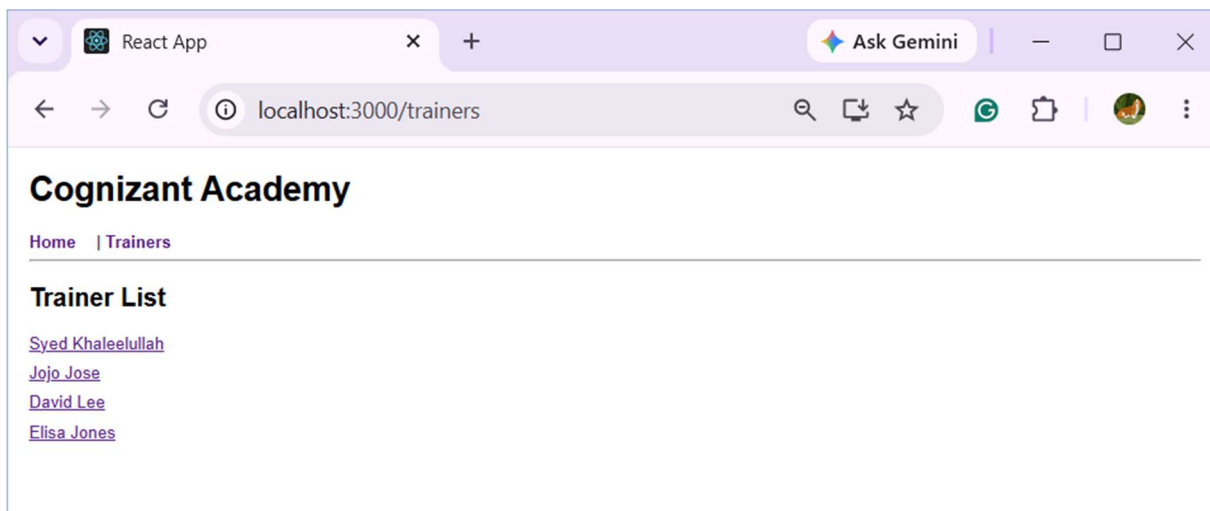
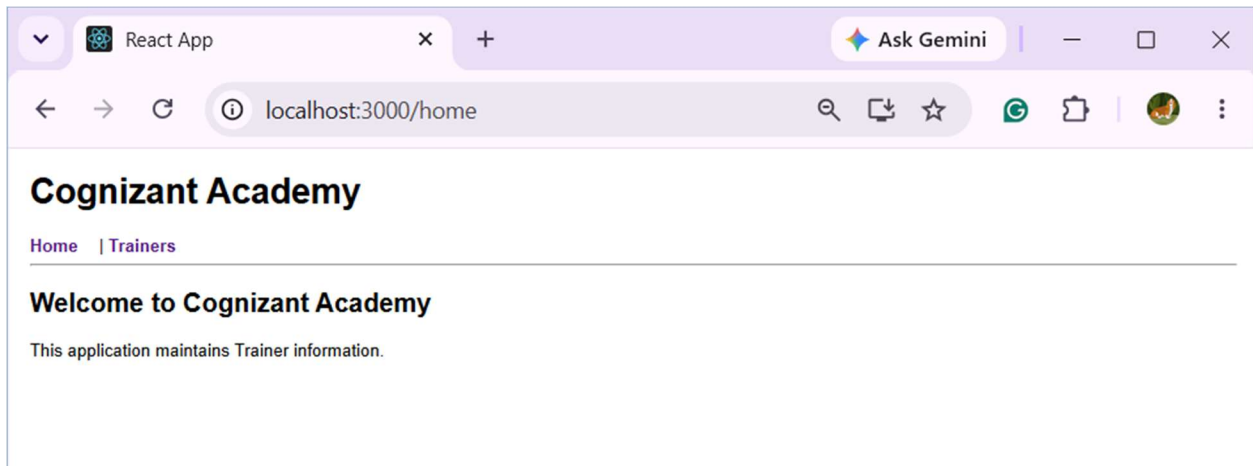
Index.css



The screenshot shows the Visual Studio Code editor with the file `index.css` open. The editor has a dark theme and a sidebar on the left with icons for Explorer, Search, Run and Debug, Extensions, and Testing. The top bar shows the file path `WEEK-6 > 06.Reactjs-hol > trainersapp > src > index.css > ...`. The code in the editor is as follows:

```
1 body {
2   font-family: Arial;
3   margin: 20px;
4 }
5
6 nav a {
7   text-decoration: none;
8   margin-right: 15px;
9   font-weight: bold;
10 }
11
12 nav a:hover {
13   color: blue;
14 }
15
16 ul {
17   list-style: none;
18   padding: 0;
19 }
20
21 li {
22   margin: 10px 0;
23 }
```

OUTPUT



React App

Ask Gemini

localhost:3000/trainer/1

Cognizant Academy

[Home](#) | [Trainers](#)

Trainer Details

ID: 1

Name: Syed Khaleelullah

Email: Khaleelullah@cognizant.com

Phone: 9876543210

Technology: Java

Skills: Spring, Hibernate, Maven, C#

React App

Ask Gemini

localhost:3000/trainer/2

Cognizant Academy

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Trainer Details

ID: 2

Name: Jojo Jose

Email: jojo@cognizant.com

Phone: 9876543211

Technology: React

Skills: ReactJS, Redux, JavaScript

React App

Ask Gemini

localhost:3000/trainer/3

Cognizant Academy

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Trainer Details

ID: 3

Name: David Lee

Email: david@gmail.com

Phone: 9876543212

Technology: Python

Skills: Python, Django, Flask

React App

Ask Gemini

localhost:3000/trainer/4

Cognizant Academy

Home | Trainers

Trainer Details

ID: 4

Name: Elisa Jones

Email: elisa@cognizant.com

Phone: 9876543213

Technology: Java

Skills: Java, JSP, Angular